

B. TECH
(SEM VII) THEORY EXAMINATION 2022-23
ENVIRONMENTAL BIOTECHNOLOGY

Time: 3 Hours

Total Marks: 100

Notes: Attempt all Sections and Assume any missing data.

Appropriate marks are allotted to each question, answer accordingly.

SECTION-A

1. Attempt all of the following Questions in brief

10x2=20

- (a) Define land pollution.
- (b) What are the toxic hydrocarbons?
- (c) What are the acetogens?
- (d) What do you understand by fermentation?
- (e) Define waste.
- (f) What is the sludge?
- (g) What are the organic contaminants?
- (h) Define Bioleaching.
- (i) What is impact analysis?
- (j) What do you understand by AQI?

SECTION-B

2. Attempt any three of the following Questions

10x3=30

- (a) What do you understand by water pollution? Describe the procedure for the measurement of water pollution. How the water pollution could be prevented?
- (b) Explain the production process of biofuel.
- (c) Briefly explain the up-flow anaerobic sludge blanket reactor.
- (d) What is biohydrogen? Describe the production process and applications of biohydrogen.
- (e) Briefly describe the Environmental Protection Act, 1986.

SECTION-C

3. Attempt any one of the following Questions

10x1=10

- (a) Write a brief note on:
 - (i) Noise pollution
 - (ii) Decay chains
- (b) What is thermal pollution? Describe different type of radiations that causes the pollutions.

4. Attempt any one of the following Questions

10x1=10

- (a) Give a brief discussion on microbiology of waste water treatments.
- (b) Describe the minimal national standards for waste disposal.

5. Attempt any one of the following Questions

10x1=10

- (a) Describe various bioreactors that are used in waste treatment.
- (b) What is the principle and working mechanism of fluidized expanded bed reactor?

6. Attempt any one of the following Questions

10x1=10

- (a) What is the bioremediation? How it is different from biomineralization?
- (b) How the organic waste could be converted into utilizable products? Explain with suitable case study.

7. Attempt any one of the following Questions

10x1=10

- (a) What are the Water Prevention and Control of Pollution Act, 1974? Give the importance of the Water Prevention and Control of Pollution Act, 1974.
- (b) Write a brief note on:
 - (i) Hazardous Wastes
 - (ii) International environmental laws